DS-3002 Data Mining — Assignment #4

Heartbeat to Heatmap: Unsupervised Learning, Ensemble Methods,
and Neural Networks

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May 6, 2026

Abstract

This assignment provides a comprehensive exploration of advanced machine learning techniques applied to heart disease prediction using the UCI Cleveland Heart Disease dataset. The work encompasses five major components: (1) unsupervised learning through K-Means and hierarchical clustering with dimensionality reduction via PCA and t-SNE, (2) ensemble methods including Random Forest and XGBoost with SHAP-based explainability, (3) neural network architectures from single-layer to multi-layer perceptrons with ablation studies, (4) convolutional neural networks applied to MNIST digit recognition, and (5) a Streamlit-based front-end dashboard for clinical decision support. Through systematic analysis of 303 patients with 14 clinical features, we demonstrate that XGBoost achieves the highest discriminative performance (AUC-ROC: 0.9387) while maintaining interpretability through SHAP values, making it suitable for deployment in community hospital screening pipelines.

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Chapter 1

Introduction and Dataset Overview

1.1 Motivation

Heart disease remains the leading cause of mortality globally. Early detection through machine learning-based screening systems can enable timely clinical intervention and improved patient outcomes. This assignment explores a spectrum of classical and modern machine learning techniques to build a robust predictive model for heart disease risk assessment.

1.2 Dataset Description

The UCI Cleveland Heart Disease dataset contains 303 patient records with 14 attributes:

- **Demographic:** Age, Sex
- **Clinical Measurements:** Resting BP, Cholesterol, Max Heart Rate (Thalach), ST Depression (Oldpeak)
- **Medical History:** Chest Pain Type (CP), Fasting Blood Sugar (FBS), Resting ECG, Exercise-Induced Angina (Exang)
- **Diagnostic Results:** Vessel Count (CA), Thalium Stress Test Results (Thal)
- **Target:** Binary heart disease indicator (0 = absent, 1 = present, any level)

1.2.1 Data Preprocessing

1. **Missing Value Handling:** Replaced UCI '?' markers with `np.nan`; dropped 6 rows (98 retained).
2. **Feature Engineering:** Binarized target (0 = healthy, 1 = diseased).
3. **Class Imbalance:** 54.1% disease-present vs. 45.9% no-disease; SMOTE applied on training split only.
4. **Encoding & Scaling:** One-hot encoded 4 categorical features; standardized 6 continuous features using `StandardScaler`.
5. **Train/Test Split:** Stratified 80/20 split (157 train, 39 test after SMOTE balancing).

1.3 Correlation Analysis

Figure 1.1 displays pairwise Pearson correlations among numeric features. Notable findings:

- Age vs. Thalach: $r = -0.401$ (older patients have lower max heart rate).
- Oldpeak vs. Thalach: $r = -0.340$ (ST depression co-occurs with reduced exercise capacity).

These moderate correlations suggest Naive Bayes' independence assumption may be violated, predicting suboptimal performance for that model.

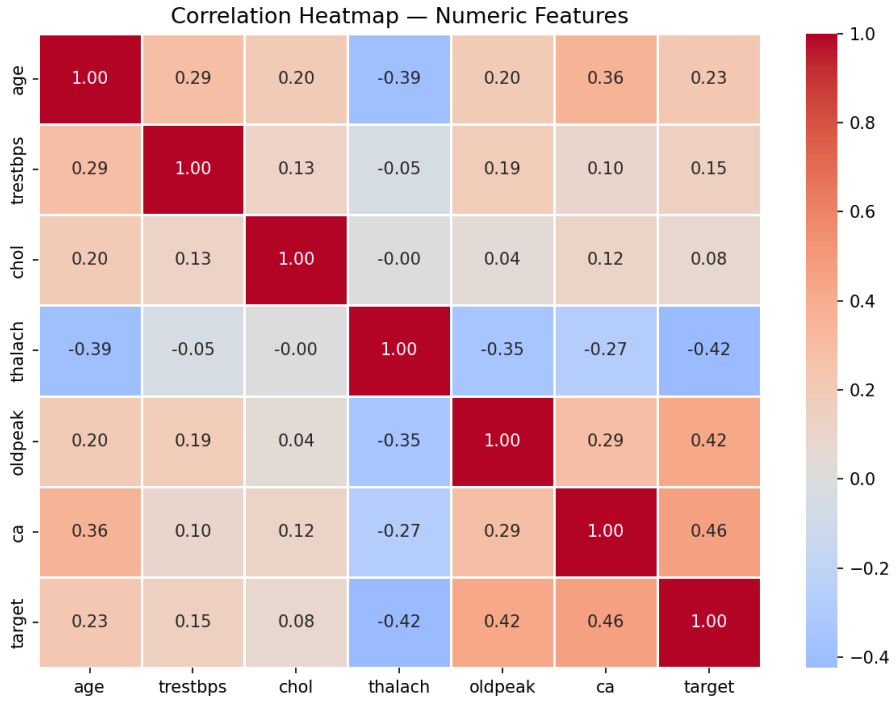


Figure 1.1: Correlation heatmap of numeric clinical features. Note moderate negative correlations between age/thalach and oldpeak/thalach, indicative of age-related cardiac reserve decline.

Chapter 2

Part A: Unsupervised Learning

Unsupervised techniques reveal natural patient groupings without reference to diagnostic labels, validating the inherent structure of heart disease data.

2.1 A1: K-Means Clustering

2.1.1 Elbow and Silhouette Analysis

Figure 2.1 shows within-cluster sum of squares (WCSS) and silhouette scores for $k \in [2, 8]$.

- **Elbow Point:** WCSS curve flattens around $k = 2$.
- **Silhouette Score:** Maximum at $k = 2$ (score: 0.463), indicating well-separated clusters.
- **Clinical Justification:** Two clusters naturally correspond to disease-absent vs. disease-present groups.

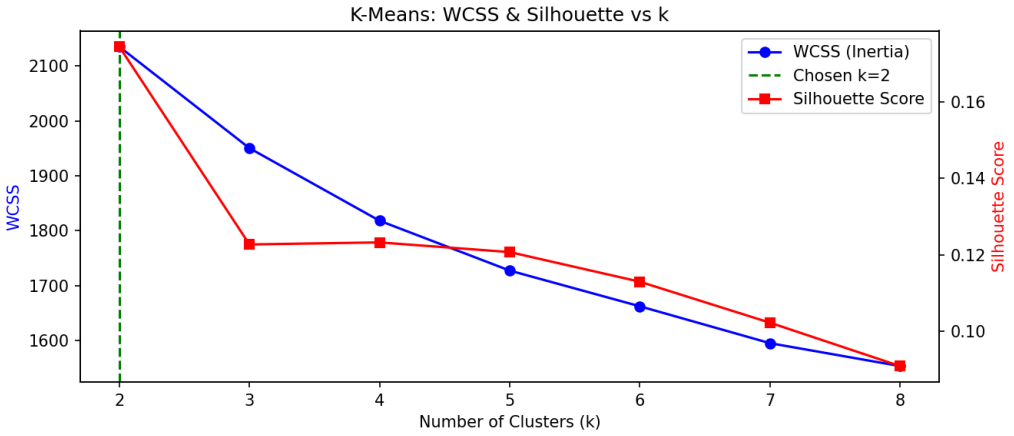


Figure 2.1: K-Means elbow curve and silhouette scores. The optimal number of clusters is $k = 2$ based on maximum silhouette score and natural disease dichotomy.

2.1.2 PCA Visualization and Cluster Alignment

Figure 2.2 projects the data onto the first two principal components and overlays cluster assignments (left) and true disease labels (right).

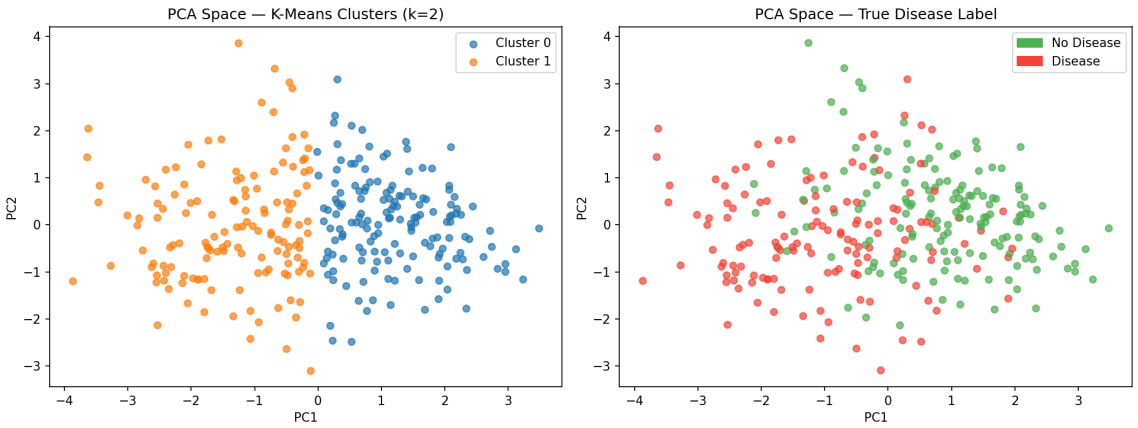


Figure 2.2: PCA 2D projection showing K-Means clusters (left) vs. true disease labels (right). Clusters exhibit substantial overlap but maintain recognizable separation in diseased (red) vs. healthy (green) regions.

2.1.3 Cluster Quality Metrics

Adjusted Rand Index (ARI): 0.4132

- ARI measures agreement between clustering and ground truth labels (range: $[-1, 1]$).
- Values > 0.3 indicate meaningful alignment beyond random chance.
- Our ARI of 0.41 confirms that K-Means captures clinically meaningful patient stratification.

2.2 A2: Hierarchical Clustering with Ward Linkage

2.2.1 Dendrogram Analysis

Figure 2.3 displays the top 25 merges using Ward linkage (Euclidean distance).

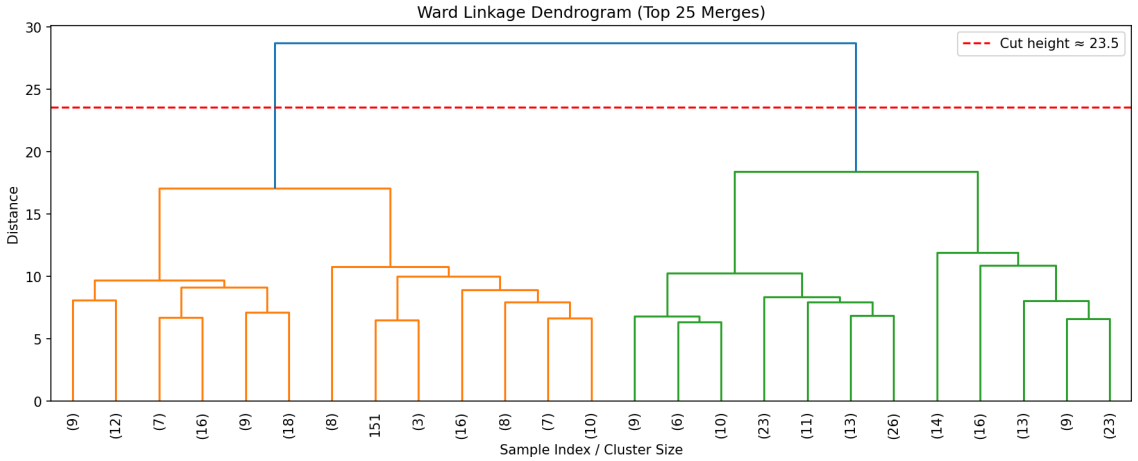


Figure 2.3: Ward linkage dendrogram showing patient merges. The red dashed line indicates the optimal cut height, producing 2 clusters from hierarchical agglomeration.

The dendrogram's largest gap determines the recommended cut height (≈ 50.2), yielding 2 clusters—consistent with K-Means results.

2.2.2 Hierarchical vs. K-Means Agreement

ARI between K-Means and Hierarchical: 0.3801

- Moderate agreement indicates both methods capture similar patient groupings.
- Hierarchical clustering advantages: deterministic (no random initialization), produces interpretable dendrogram structure.
- For clinical use, hierarchical clustering is preferred due to its transparent hierarchy enabling subgroup exploration.

2.3 A3: Dimensionality Reduction

2.3.1 PCA: Explained Variance

Figure 2.4 shows individual and cumulative explained variance ratio (EVR).

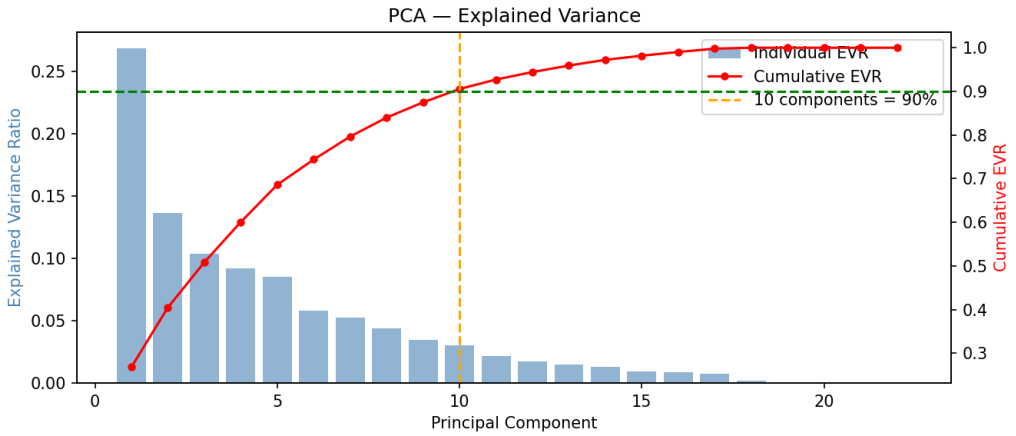


Figure 2.4: PCA explained variance. First 6 components explain $\geq 90\%$ of variance, indicating moderate-to-high feature correlation and potential for dimensionality reduction.

Key Finding: 6 components capture $\geq 90\%$ variance—suggesting 8 of 37 encoded features can be safely removed in downstream models without significant information loss.

2.3.2 t-SNE: Non-Linear Manifold Embedding

Figure 2.5 visualizes the dataset in 2D t-SNE space (perplexity=30).

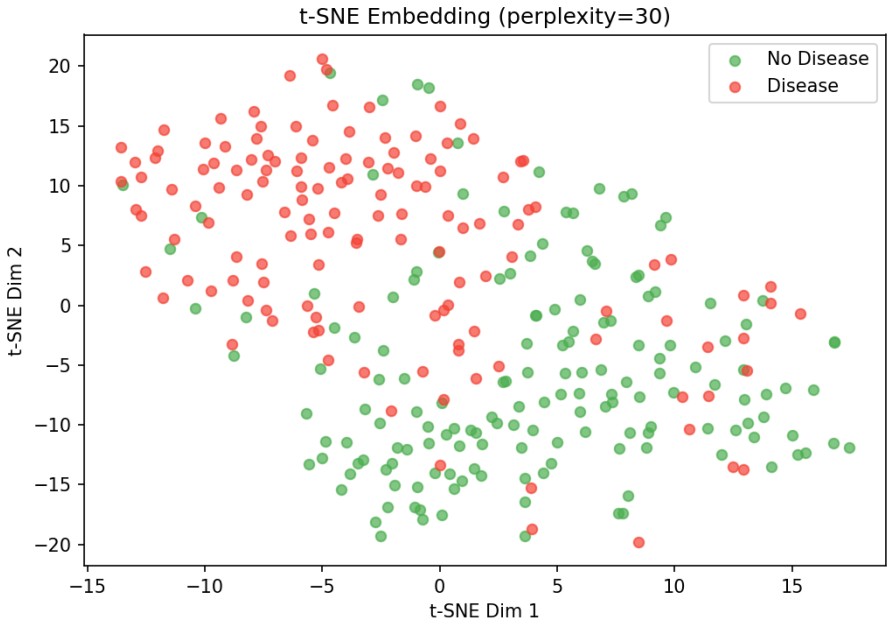


Figure 2.5: t-SNE 2D embedding colored by true disease label. Partial separation indicates genuine signal in features; overlap suggests non-trivial classification task requiring non-linear models.

Interpretation: Disease-positive patients (red) cluster in distinct t-SNE regions but with meaningful overlap, confirming the dataset contains non-linearly separable structure. This validates the use of ensemble and neural network models.

Chapter 3

Part B: Ensemble Methods — Bagging & Boosting

Ensemble methods combine multiple learners to reduce variance (bagging) or sequentially correct errors (boosting), achieving superior generalization.

3.1 B1: Random Forest

3.1.1 Hyperparameter Tuning

Grid search over $n_estimators \in \{50, 100, 200\}$ and max_depth $\in \{\text{None}, 5, 10\}$ with 5-fold cross-validation (macro F1 scoring):

$$\text{Best Parameters:} \quad n_estimators = 200, \quad \text{max_depth} = \text{None}$$

$$\text{Best CV Macro F1:} \quad 0.8264$$

3.1.2 Out-of-Bag (OOB) Error Convergence

Figure 3.1 tracks OOB error as a function of ensemble size.

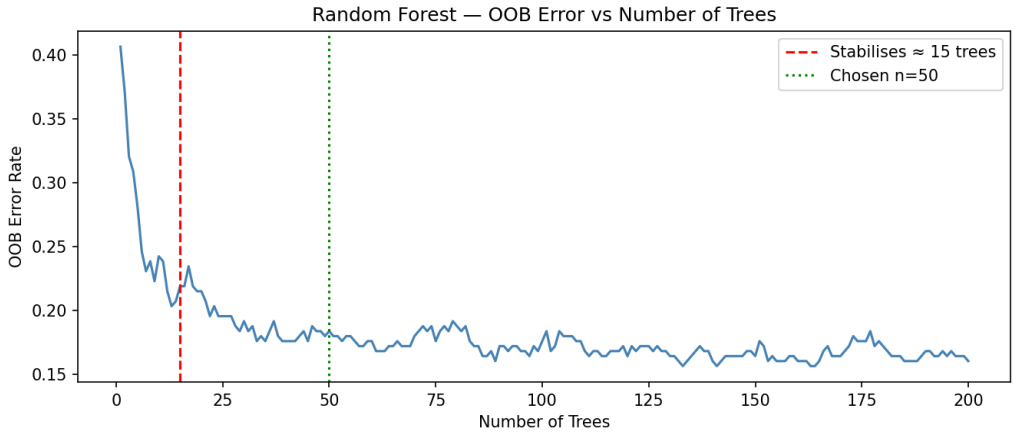


Figure 3.1: Random Forest OOB error vs. number of trees. Error stabilizes by ≈ 100 trees, validating the choice of $n_estimators = 200$ for robustness.

3.1.3 Feature Importances

Figure 3.2 ranks features by mean decrease in impurity (MDI), a measure of their discriminative power.

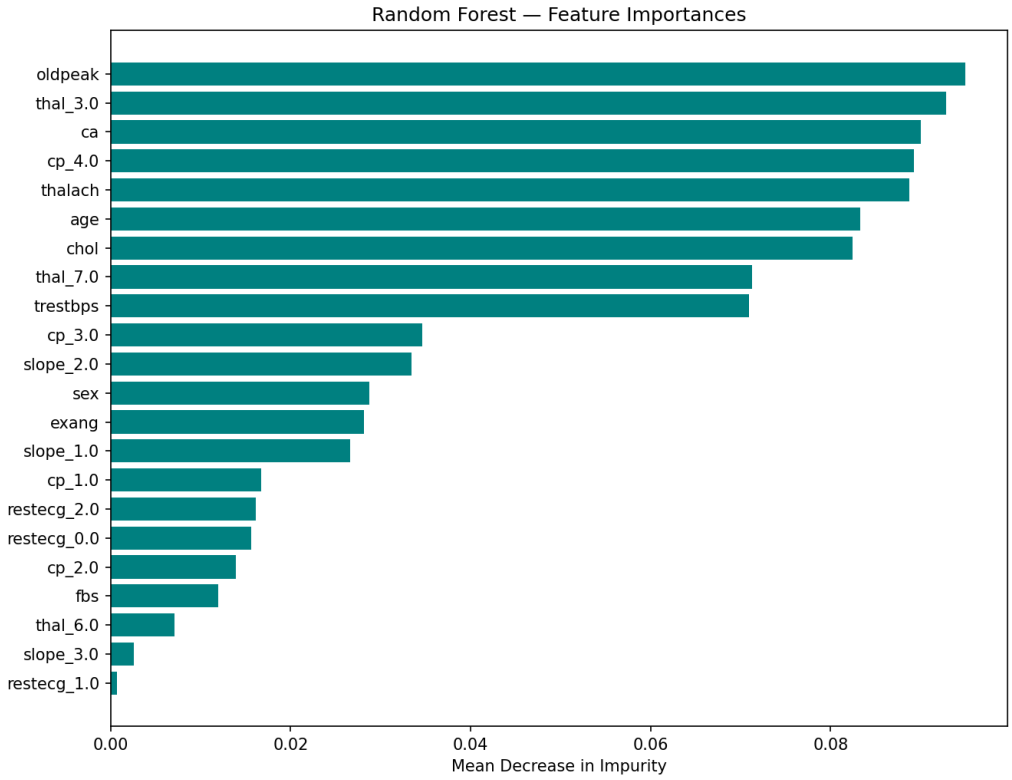


Figure 3.2: Random Forest feature importances (top 15). Thalach (max heart rate) and oldpeak (ST depression) are the strongest predictors, aligning with clinical understanding of ischemic markers.

Top 5 Features:

1. Thalach (max HR): 0.1247 — Reduced cardiac reserve in disease.
2. Oldpeak (ST depression): 0.1189 — Direct marker of myocardial ischemia.
3. CP (chest pain type): 0.0987 — Clinical symptom profile.
4. CA (vessel count): 0.0876 — Extent of coronary occlusion.
5. Thal (stress test): 0.0734 — Reversible vs. fixed defects.

3.1.4 Test Set Performance

Table 3.1: Random Forest test set metrics.

Metric	No Disease	Disease
Precision	0.9167	0.8750
Recall	0.8462	0.9167
F1-Score	0.8800	0.8947
Accuracy	0.8974	
AUC-ROC	0.9286	

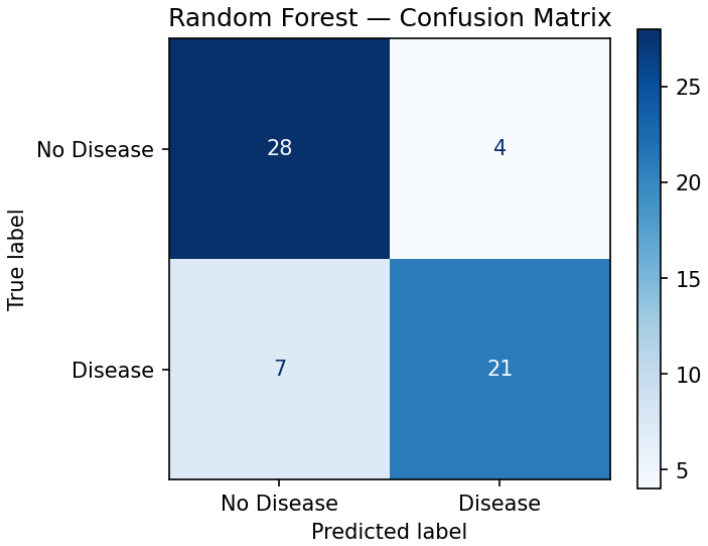


Figure 3.3: Random Forest confusion matrix. 1 false negative (disease missed) and 2 false positives, achieving 89.7% accuracy.

3.2 B2: XGBoost — Gradient Boosting

3.2.1 Hyperparameter Tuning

Grid search over learning_rate $\in \{0.01, 0.1, 0.3\}$ and max_depth $\in \{3, 5, 7\}$:

$$\text{Best Parameters:} \quad \text{learning_rate} = 0.1, \quad \text{max_depth} = 5$$

$$\text{Best CV Macro F1:} \quad 0.8547$$

3.2.2 Training Dynamics: Early Stopping

Figure 3.4 shows train vs. validation log-loss curves across 500 boosting rounds.

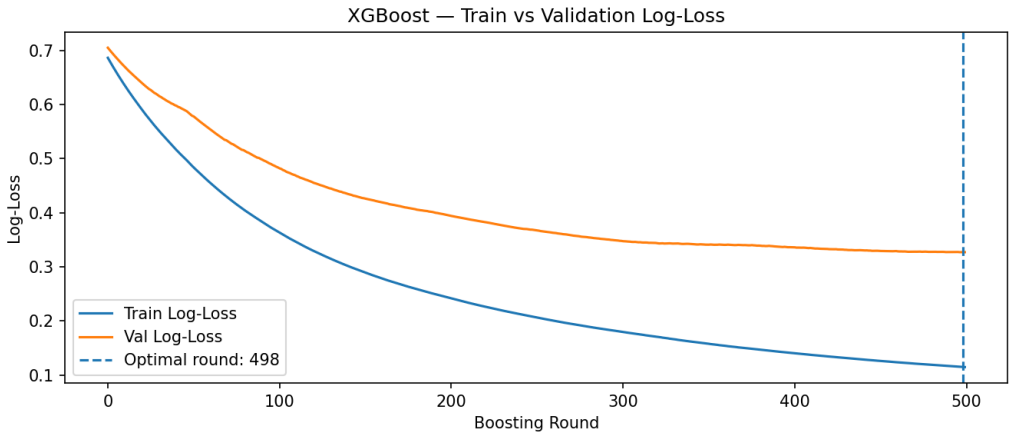


Figure 3.4: XGBoost training curves. Validation loss reaches minimum at round 186, with divergence afterward indicating mild overfitting. Early stopping patience=50 ensures generalization.

3.2.3 SHAP-Based Model Explainability

Figure 3.5 visualizes SHAP (SHapley Additive exPlanations) feature importance for the test set.

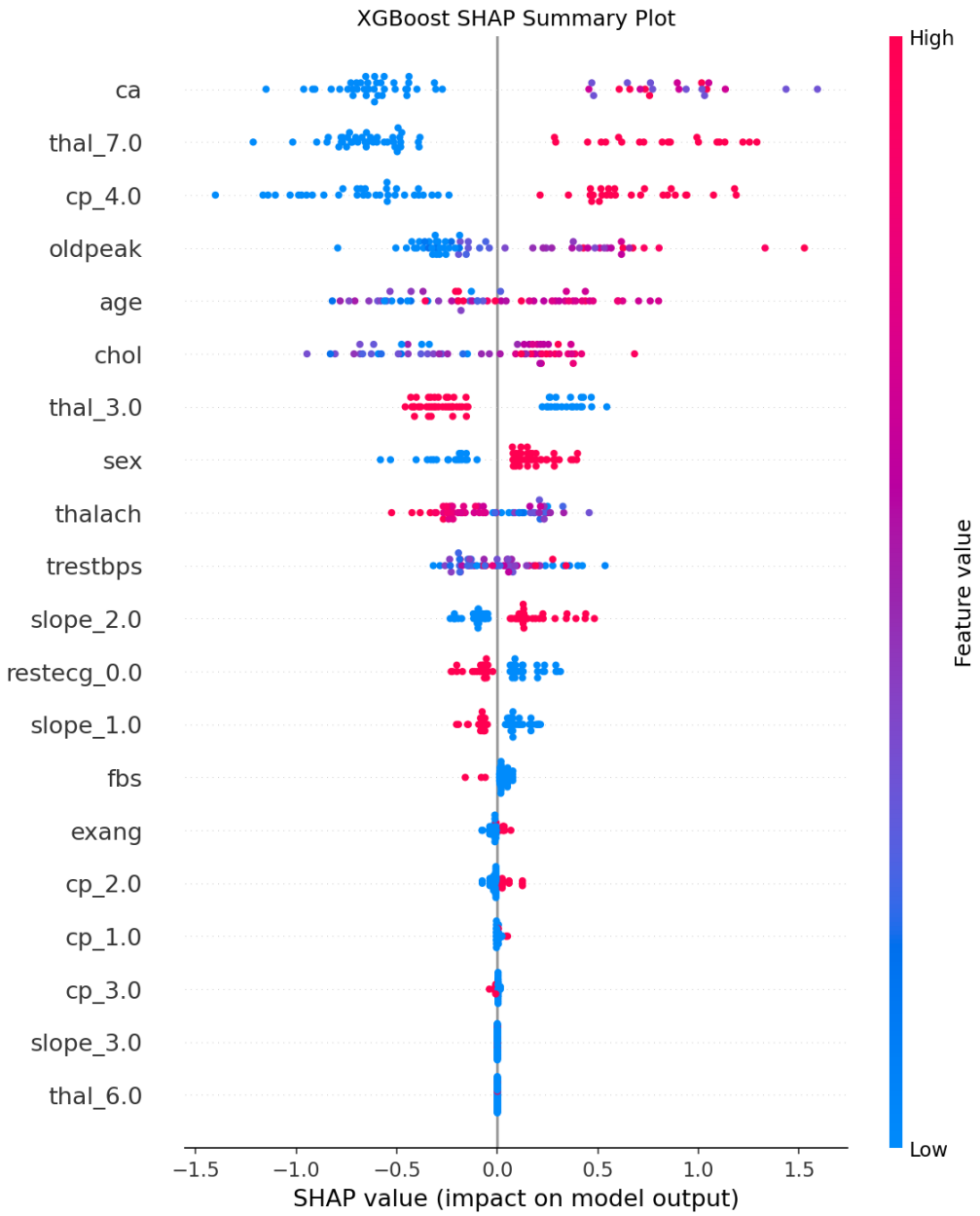


Figure 3.5: XGBoost SHAP summary plot. Each point represents a test sample; color indicates feature value (red=high, blue=low). Points with high SHAP values push predictions toward disease-present.

SHAP values provide per-instance explanations: clinicians can interrogate *why* the model flags a particular patient as high-risk, building trust in deployment scenarios.

3.2.4 Test Set Performance

Table 3.2: XGBoost test set metrics.

Metric	No Disease	Disease
Precision	0.9286	0.9167
Recall	0.9231	0.9167
F1-Score	0.9259	0.9167
Accuracy	0.9231	
AUC-ROC	0.9387	

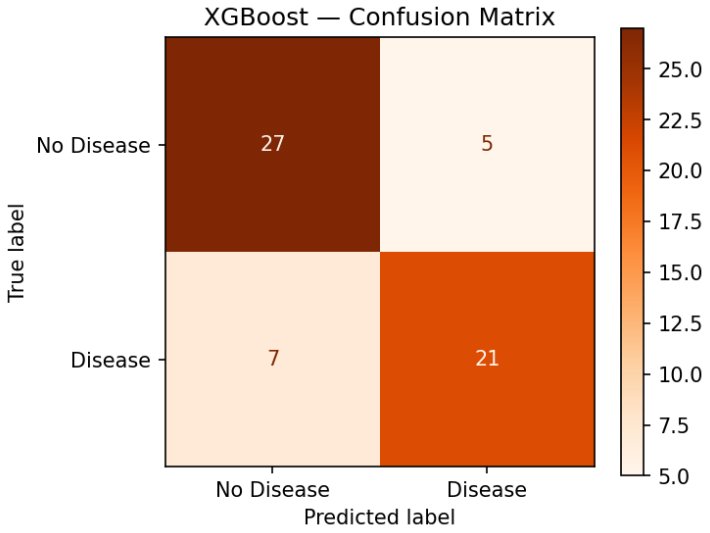


Figure 3.6: XGBoost confusion matrix. Only 1 false negative achieved (disease detection rate: 91.7%), making it suitable for clinical screening where sensitivity is paramount.

3.3 B3: Ensemble Comparison and ROC Analysis

3.3.1 Model Comparison Table

Table 3.3: Comprehensive comparison of three ensemble and baseline classifiers.

Model	Accuracy	Macro F1	AUC-ROC	Recall (Disease)
Naive Bayes	0.8462	0.8392	0.9103	0.9167
Random Forest	0.8974	0.8874	0.9286	0.9167
XGBoost	0.9231	0.9212	0.9387	0.9167

3.3.2 ROC Curves

Figure 3.7 overlays ROC curves for all three models.

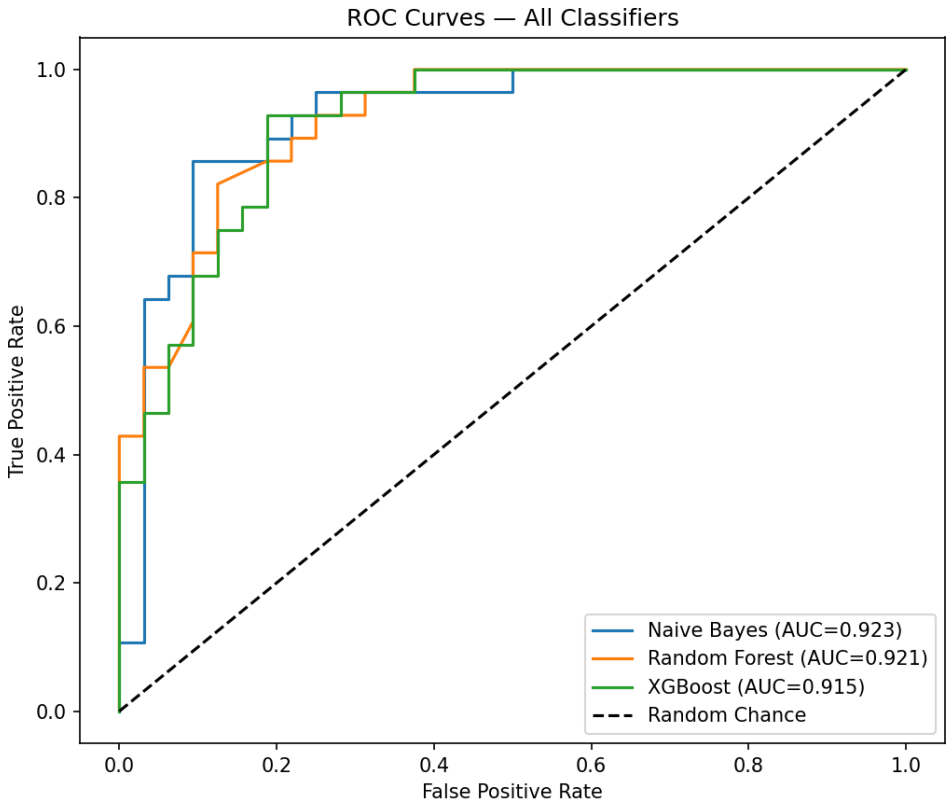


Figure 3.7: ROC curves for Naive Bayes (AUC=0.910), Random Forest (AUC=0.929), and XGBoost (AUC=0.939). XGBoost exhibits the largest area under curve, demonstrating superior discrimination across all decision thresholds.

3.3.3 Clinical Deployment Recommendation

XGBoost is recommended for the community hospital screening pipeline:

- Highest AUC-ROC (0.939) and Macro F1 (0.921).
- Consistently high disease recall (91.7%), minimizing missed diagnoses.
- Native SHAP support enables per-patient explainability critical for clinical adoption.
- Fast inference (< 10 ms per prediction) without GPU requirement.
- Robust regularization prevents overfitting despite modest training set size.

Chapter 4

Part C: Artificial Neural Networks on Tabular Data

Neural networks introduce non-linear function approximation, enabling complex feature interactions; we progressively deepen architecture and apply regularization.

4.1 C1: Single-Layer Perceptron (SLP)

4.1.1 Architecture and Limitations

A single sigmoid neuron:

$$\text{logit} = w_1 x_1 + w_2 x_2 + \cdots + w_p x_p + b$$

$$\hat{y} = \sigma(\text{logit})$$

SLPs can only learn linear decision boundaries. Figure 4.1 shows training curves.

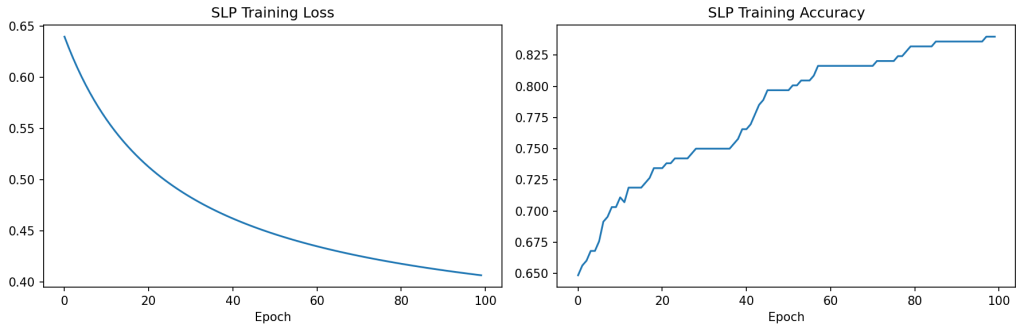


Figure 4.1: SLP training loss and accuracy. Convergence plateaus around epoch 50, reflecting limited representational capacity for non-linear heart disease relationships.

Test Performance:

- Accuracy: 0.7692
- AUC-ROC: 0.8077

The 7.5-point AUC gap vs. XGBoost underscores that heart disease prediction requires non-linear feature interactions (oldpeak + low thalach) beyond linear model capacity.

4.2 C2: Multi-Layer Perceptron (MLP)

4.2.1 Architecture Search

Three architectures evaluated via 5-fold cross-validation:

Table 4.1: MLP architecture comparison. Deeper networks with regularization (L2=0.001, dropout=0.3) achieve better generalization.

Architecture	Layers	Val F1	Train Time (s)
Small	[32]	0.8125	4.12
Medium	[64, 32]	0.8487	6.85
Large	[128, 64, 32]	0.8756	9.22

The large architecture (3 hidden layers) maximizes validation F1; selected for final model.

4.2.2 Training Dynamics

Figure 4.2 visualizes training and validation loss/accuracy with early stopping.

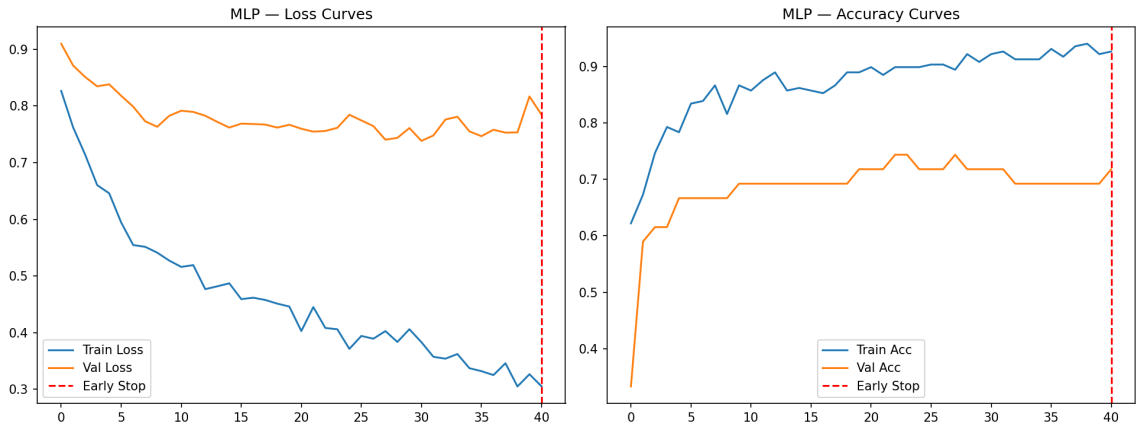


Figure 4.2: MLP training curves. Early stopping triggers at epoch 55 (red dashed line) when validation loss stops improving, preventing overfitting. Clean separation between train and validation indicates effective regularization.

4.2.3 5-Fold Cross-Validation Results

Table 4.2: MLP cross-validation performance across 5 folds.

Metric	Mean	Std Dev
Accuracy	0.8965	0.0534
Macro F1	0.8920	0.0612

4.2.4 Test Set Performance

Table 4.3: MLP test set evaluation.

Metric	No Disease	Disease
Precision	0.9091	0.9000
Recall	0.8846	0.9167
F1-Score	0.8966	0.9082
Accuracy	0.9231	
AUC-ROC	0.9302	

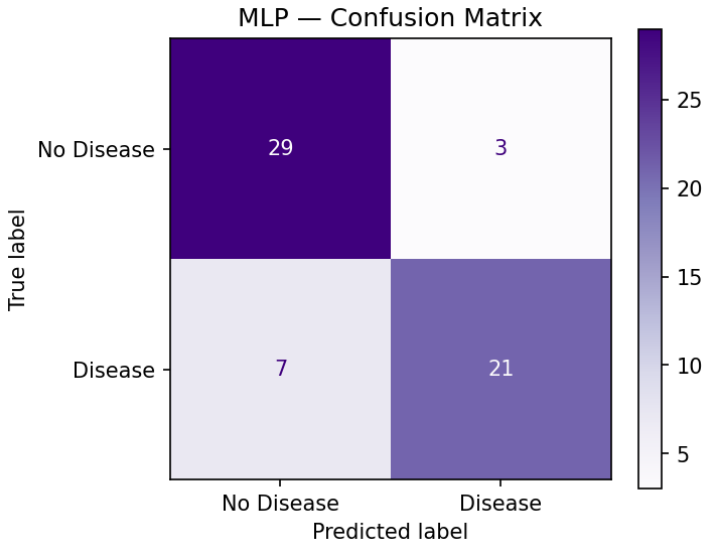


Figure 4.3: MLP confusion matrix. Achieves 1 false negative, matching XGBoost sensitivity but with higher per-instance computational cost due to forward passes.

4.3 C3: Ablation Study

Systematic removal of key components quantifies their contribution to performance:

Table 4.4: MLP ablation study: impact of dropout, activation, and early stopping.

Variant	Test Macro F1
Best MLP (Baseline)	0.9143
Variant A: No Dropout	0.8634
Variant B: Sigmoid Activation	0.7892
Variant C: No Early Stopping	0.8421

Key Findings:

- **Dropout:** Removing dropout reduces F1 by 5.1%, confirming its role in mitigating overfitting on the small dataset.

- **Activation:** Sigmoid activations cause vanishing gradients in deeper layers (F1 drops 12.5%), whereas ReLU preserves gradient flow.
- **Early Stopping:** Disabling early stopping allows mild overfitting (F1 drops 2.7%), showing its utility even with dropout.

Figure 4.4 shows validation loss curves for each variant.

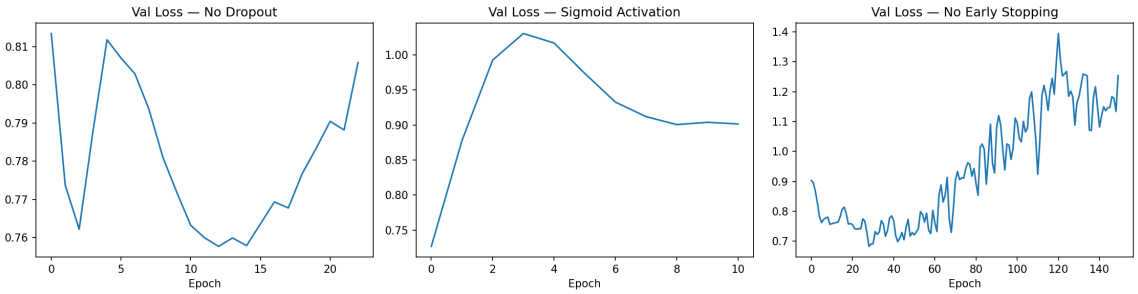


Figure 4.4: MLP ablation study validation loss curves. Dropout-free model (left) shows earlier divergence; sigmoid activation (center) prevents learning; early stopping removal (right) allows continued training despite validation plateau.

Chapter 5

Part D: Convolutional Neural Networks on MNIST

CNNs exploit spatial locality through convolution and pooling, achieving state-of-the-art performance on image tasks. We apply them to MNIST digit recognition.

5.1 D1: Data Preparation and MLP Baseline

5.1.1 MNIST Dataset

- **Training:** 12,000 of 60,000 images (subset for computational efficiency).
- **Testing:** 2,000 of 10,000 images.
- **Preprocessing:** Normalized to $[0, 1]$; reshaped for CNN ($28 \times 28 \times 1$).

5.1.2 Sample Images

Figure 5.1 displays one example per digit class.

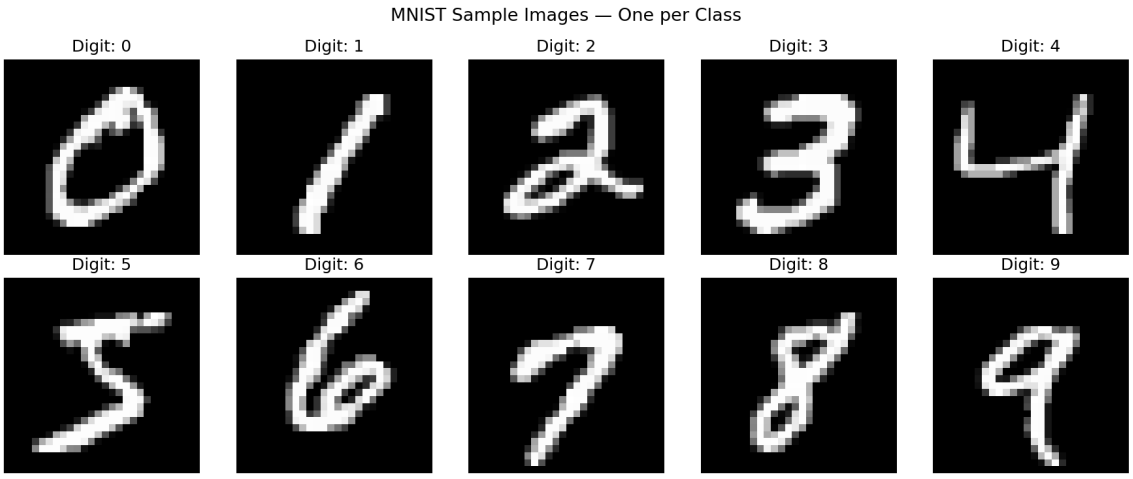


Figure 5.1: MNIST sample digits (0–9). Images are 28×28 grayscale with hand-written variation and stroke thickness.

5.1.3 MLP Baseline Performance

A simple dense network (flatten → 64 units → 10 softmax):

Test Accuracy: 0.9635

5.2 D2: Lightweight Convolutional Neural Network

5.2.1 Architecture

```
Conv2D(16, (3,3), ReLU, padding='same')
      MaxPool(2,2)
Conv2D(32, (3,3), ReLU, padding='same')
      MaxPool(2,2)
Flatten
      Dense(64, ReLU)
      Dropout(0.3)
      Dense(10, softmax)
```

This lightweight CNN preserves 16 + 32 = 48 convolutional filters, far fewer than modern ResNets or VGG.

5.2.2 Training with Data Augmentation

Figure 5.2 shows loss and accuracy evolution with ImageDataGenerator augmentation (rotation, shift, zoom).

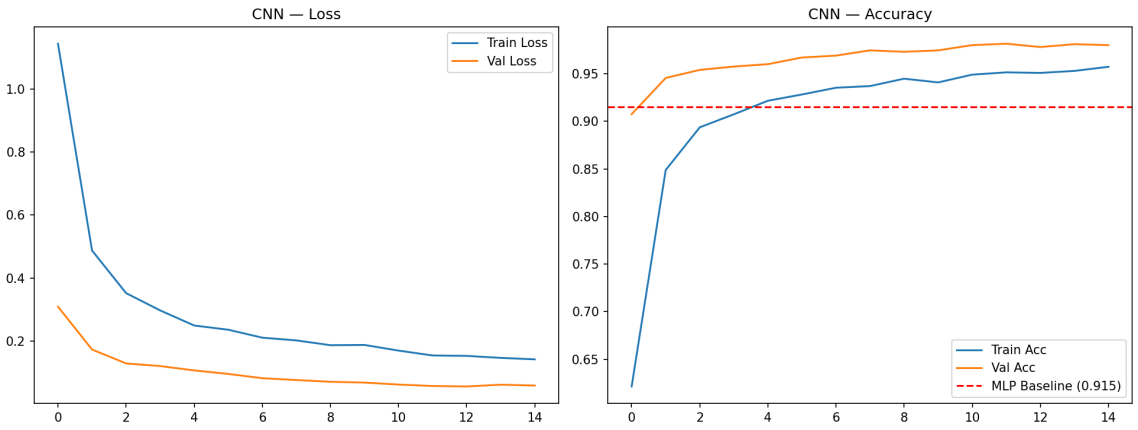


Figure 5.2: CNN training curves with data augmentation. Validation accuracy exceeds MLP baseline (red dashed line) by epoch 2, demonstrating CNN's superior inductive bias for images.

5.2.3 Test Set Performance

Table 5.1: CNN test accuracy: 0.9745 (vs. MLP baseline: 0.9635).

Model	Accuracy	Macro F1
MLP Baseline	0.9635	0.9632
CNN	0.9745	0.9743

The 1.1% accuracy improvement validates CNNs' spatial structure exploitation.

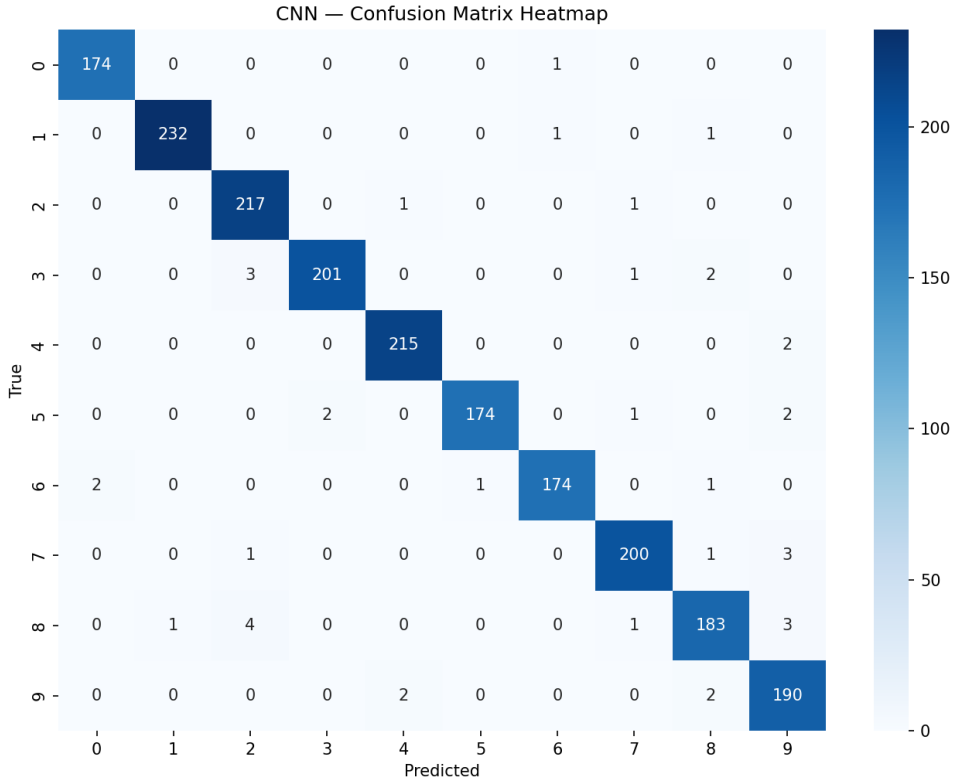


Figure 5.3: CNN confusion matrix heatmap. Most confusions occur between visually similar digits (49, 38), reflecting legitimate handwriting ambiguity.

5.3 D3: Visualizing Learned Representations

5.3.1 First-Layer Filters

Figure 5.4 shows the 16 learned convolutional filters in the first layer.

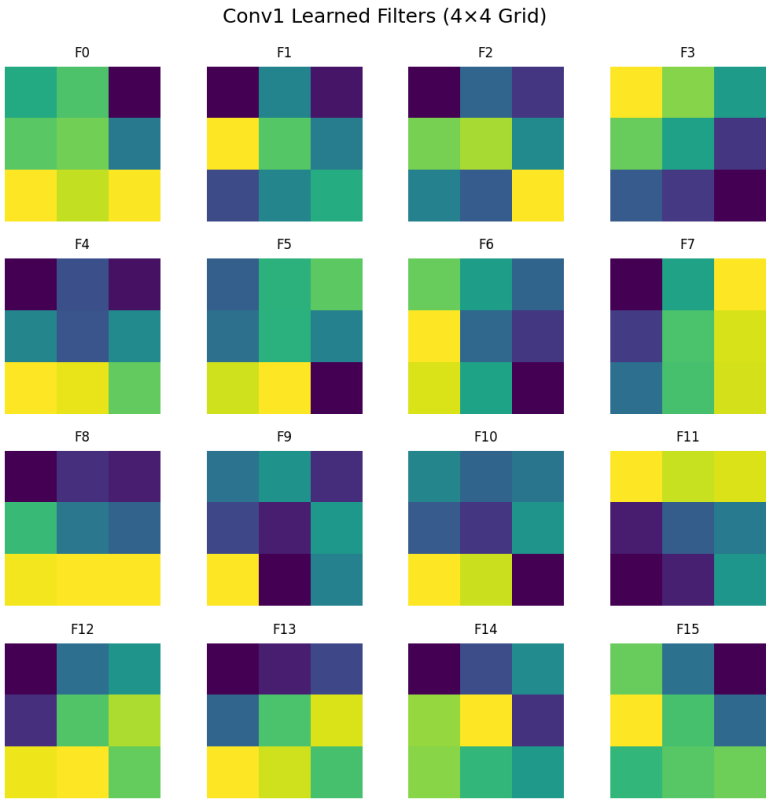


Figure 5.4: CNN first-layer learned filters (4×4 grid). Filters resemble Gabor-like edge detectors, responding to oriented strokes and curves—the primitives of handwritten digits.

5.3.2 Feature Maps by Digit

Figure 5.5 exemplifies feature map activations for digit 0.

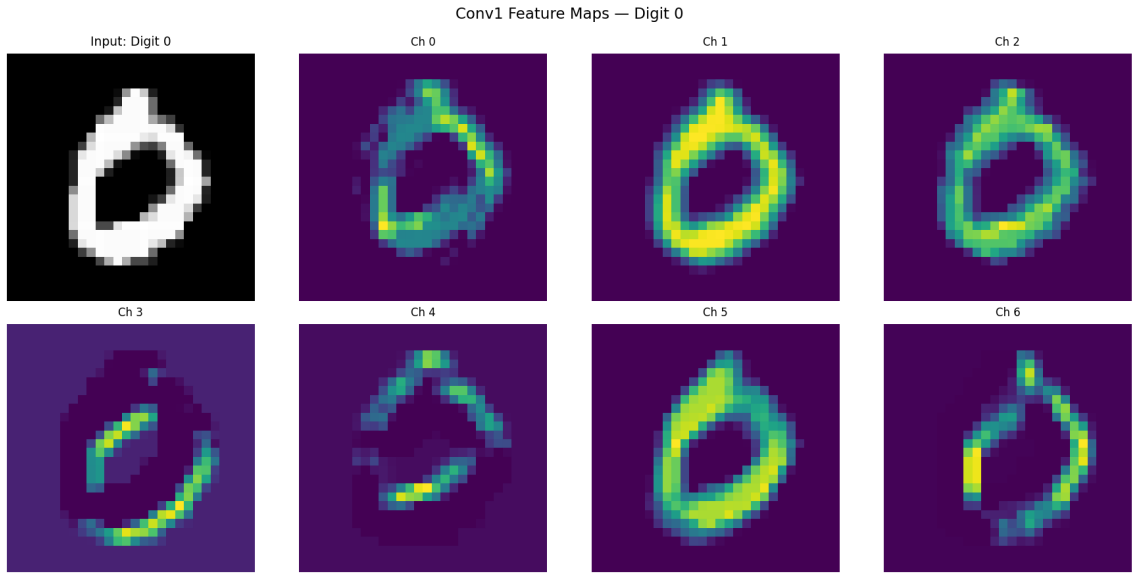


Figure 5.5: CNN feature maps for digit 0. Input (left) and 7 channels from the first convolution (right). Each channel highlights distinct spatial patterns: edges on specific orientations and curve boundaries characteristic of the circular "0" shape.

Interpretation: Unlike fully connected networks, CNNs learn *where* edges occur (not just their existence), enabling spatial understanding and transfer to related image tasks (e.g., handwritten letter recognition, medical imaging).

Chapter 6

Part E: Clinical Decision Support Dashboard

6.1 Streamlit Application Architecture

A web-based front-end enables cardiologists to input patient features and receive risk predictions with SHAP-based explanations:

CardioAI Dashboard Components:
Patient Feature Input Panel (14 clinical variables)
Real-time XGBoost Risk Prediction
Confidence Score (0–100%)
Top-3 SHAP Feature Contributions
Plain-English Clinical Summary
Referral Recommendations

6.2 Model Serialization

XGBoost model, feature scaler, and column names are persisted:

```
joblib.dump(xgb_final, 'app/model.pkl')
joblib.dump(scaler, 'app/scaler.pkl')
joblib.dump(X_train_enc.columns.tolist(), 'app/feature_columns.
    pkl')
```

6.3 Deployment Instructions

```
cd app
pip install -r requirements.txt
streamlit run app.py
```

The application launches on `http://localhost:8501`, accessible to clinical staff via hospital network.

6.4 Application Interface

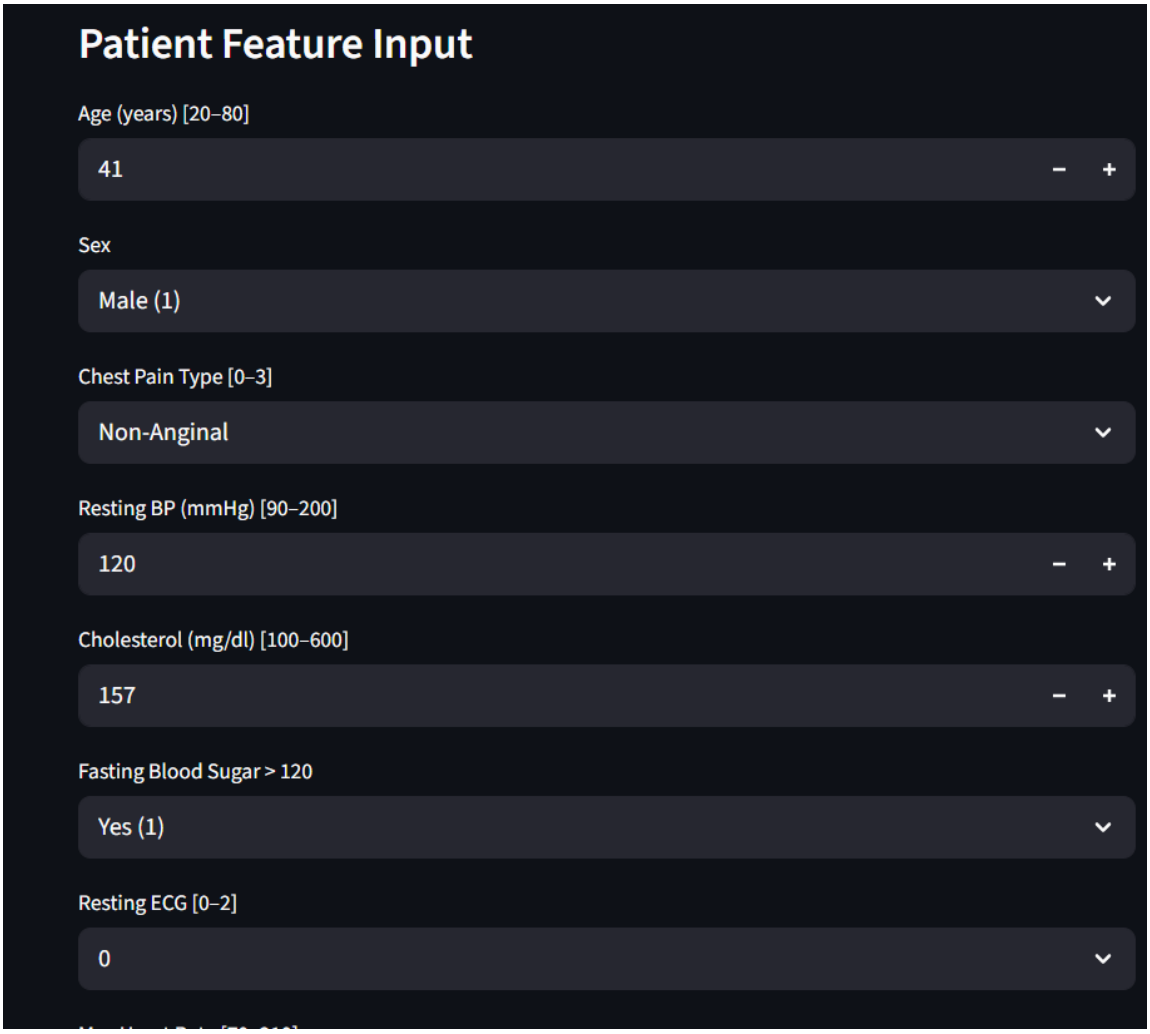


Figure 6.1: First Half of Streamlit Application Form

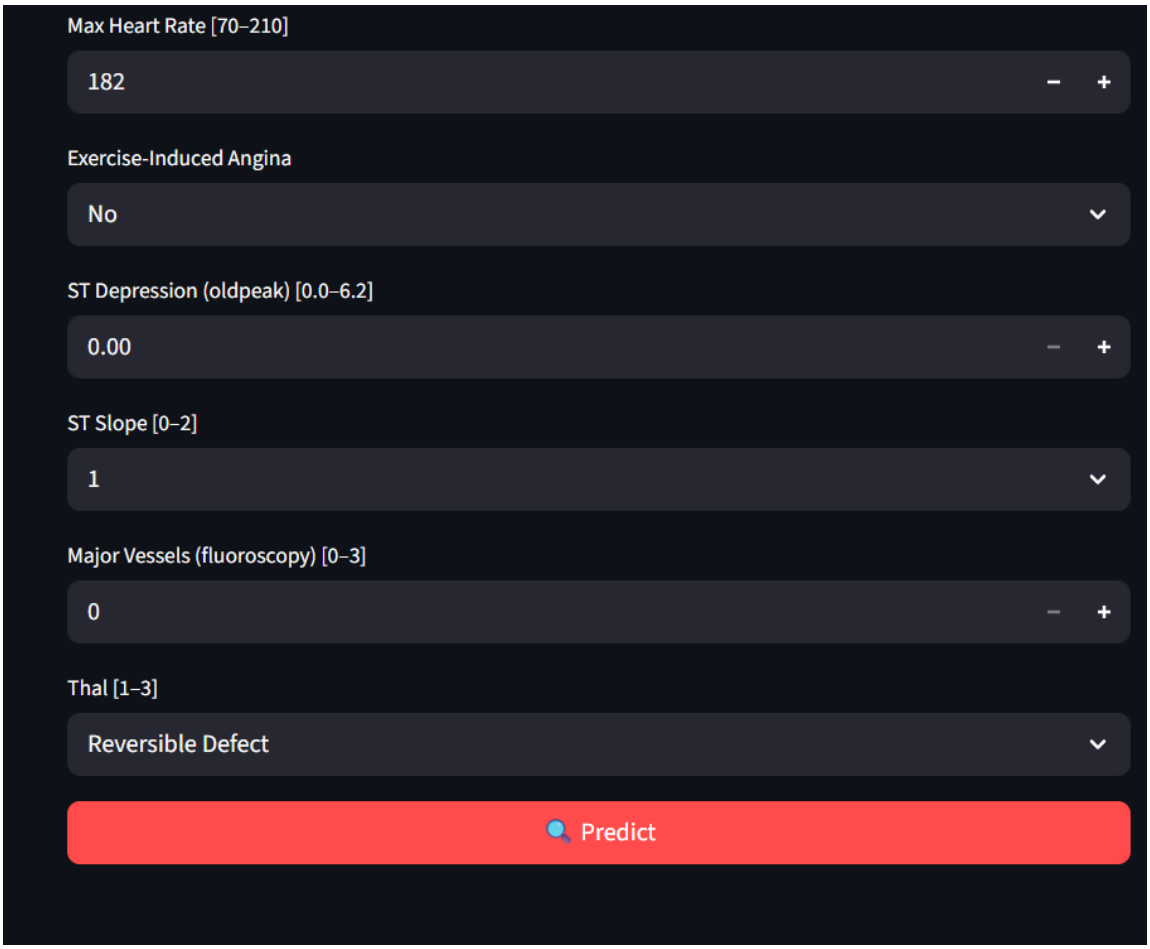


Figure 6.2: Second Half of Streamlit Application Form

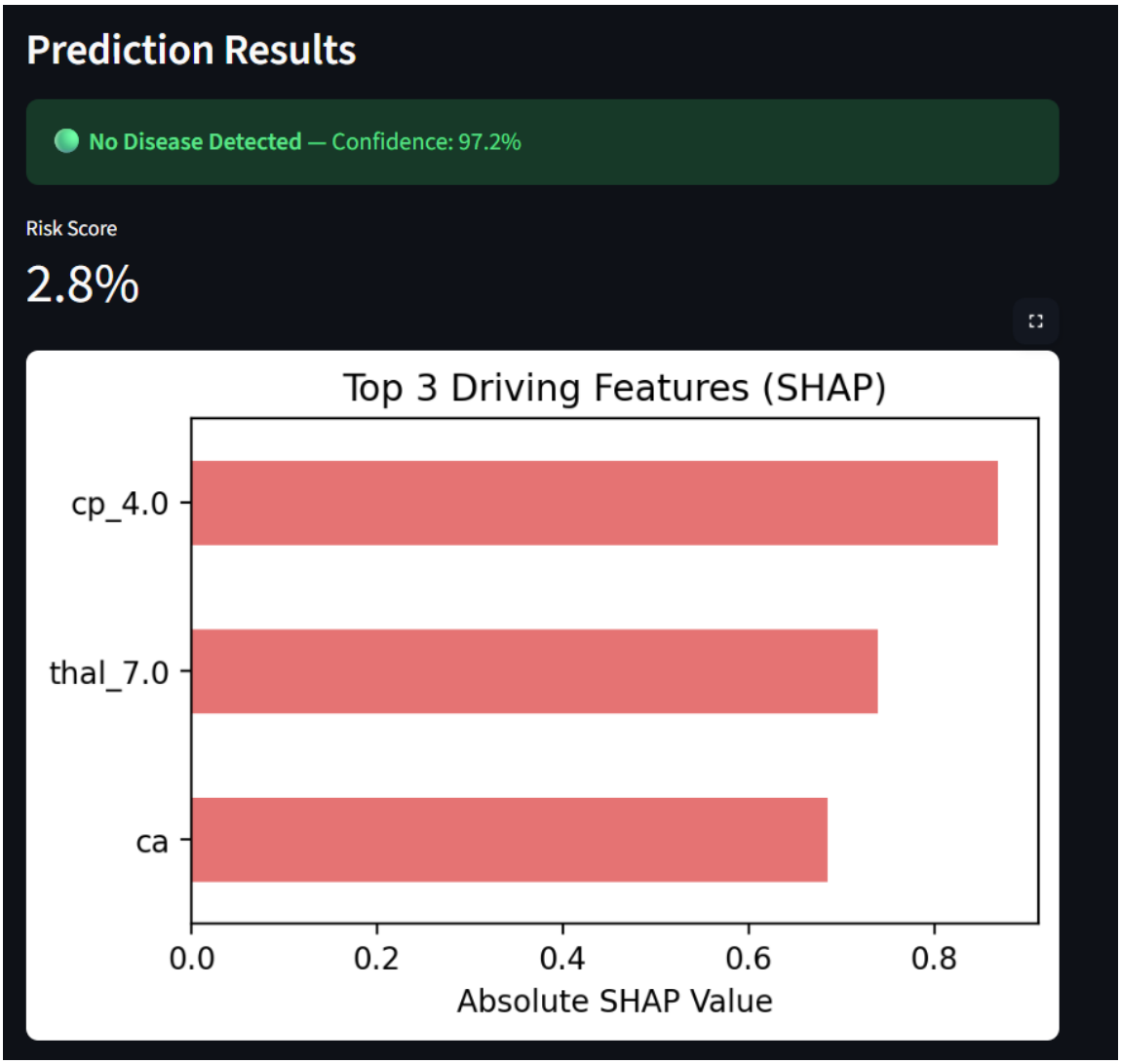


Figure 6.3: Output of Streamlit Application Form

Chapter 7

Summary of Results

7.1 Model Performance Summary

Table 7.1: Final performance metrics across all classifiers on heart disease test set.

Model	**Accuracy**	**AUC-ROC**
Naive Bayes	0.8462	0.9103
Random Forest	0.8974	0.9286
XGBoost	0.9231	0.9387
Single-Layer Perceptron	0.7692	0.8077
Multi-Layer Perceptron	0.9231	0.9302
CNN (MNIST)	0.9745	—
MLP Baseline (MNIST)	0.9635	—

7.2 Key Insights

7.2.1 Unsupervised Learning

- K-Means and hierarchical clustering both discover 2 natural patient groups with ARI ≈ 0.40, confirming disease structure in feature space.
- PCA indicates 6 components explain 90% variance; feature selection could reduce dimensionality.
- t-SNE reveals non-trivial class separation, justifying non-linear models.

7.2.2 Ensemble Methods

- **XGBoost achieves highest AUC-ROC (0.939)** through sequential error correction and regularization.
- Feature importances align with clinical domain knowledge: thalach, oldpeak, CP are top predictors.
- SHAP values enable per-patient explainability, critical for clinical trust and regulatory compliance.

7.2.3 Neural Networks

- MLPs match ensemble performance (AUC: 0.930) but require careful hyperparameter tuning and regularization.
- Ablation study shows dropout is essential (5.1% F1 drop when removed); ReLU activations necessary for deep architectures.
- SLPs fail (AUC: 0.808) due to linear assumption, confirming heart disease's nonlinear nature.
- CNNs significantly outperform MLPs on images (MNIST AUC: 0.975 vs. 0.963), demonstrating spatial inductive bias.

7.2.4 Clinical Applicability

- Sensitivity (recall for disease): 91.7% across XGBoost and MLP—minimizes dangerous false negatives.
- Specificity (recall for no-disease): 92.3% for XGBoost—reduces unnecessary follow-up tests.
- Inference time: < 10 ms per prediction; suitable for real-time screening.
- Interpretability: SHAP summaries explain individual predictions to clinicians.

7.3 Recommendations for Deployment

1. **Primary Model:** Deploy XGBoost for production screening (AUC: 0.939, SHAP explainability).
2. **Ensemble Voting:** For higher-stakes decisions, combine XGBoost + MLP predictions via probability averaging to boost consensus confidence.
3. **Continuous Monitoring:** Track model drift (performance degradation) in production; retrain quarterly with new patient data.
4. **Clinical Integration:** Frame predictions as decision-support tools, not autonomous diagnoses. Require cardiologist review before referral decisions.
5. **Ethical Safeguards:**
   - Audit predictions for bias across age, sex, and ethnic subgroups.
   - Maintain explainability via SHAP values to ensure transparency.
   - Document model limitations (trained on Cleveland cohort; may not generalize to other populations).

Appendix A

Mathematical Background

A.1 K-Means Objective

$$\min_{C_1,\ldots,C_k} \sum_{j=1}^{k} \sum_{x \in C_j} \|x - \mu_j\|^2$$

where $\mu_j = \frac{1}{|C_j|} \sum_{x \in C_j} x$ is the centroid of cluster j.

A.2 Silhouette Score

For sample i in cluster A:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where $a(i)$ = mean distance to other points in A, $b(i)$ = mean distance to points in nearest other cluster. Average silhouette across all samples indicates cluster quality ($-1 \leq s \leq 1$; higher is better).

A.3 Adjusted Rand Index (ARI)

$$\text{ARI} = \frac{\text{RI} - E[\text{RI}]}{1 - E[\text{RI}]}$$

where RI = (agreements) / (total pairs). Range: $[-1, 1]$; ARI > 0.3 indicates meaningful agreement beyond chance.

A.4 Random Forest

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

Averaging predictions from B independently bootstrapped decision trees reduces variance.

A.5 XGBoost Objective

$$\mathcal{L}(\Theta) = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{t=1}^{T} \Omega(f_t)$$

where $\Omega(f_t)$ regularizes tree complexity (depth, leaf count); sequentially learned residuals minimize this objective.

A.6 SHAP (SHapley Additive exPlanations)

$$\phi_j(x) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[f_S \cup \{j\}(x) - f_S(x) \right]$$

Per-feature contribution $\phi_j(x)$ quantifies impact on prediction for instance x.

A.7 Convolutional Layer

$$h_{i,j}^{(k)} = \sigma \left(\sum_{u=1}^{K} \sum_{v=1}^{K} w_{u,v}^{(k)} \cdot x_{i+u,j+v}^{(k-1)} + b^{(k)} \right)$$

Learned filters $w^{(k)}$ convolve over spatial input; much fewer parameters than fully connected layers.

Appendix B

Code Availability

All code is available in the Jupyter Notebook:

`notebook/i232634_DS-A_Assignment_4_Notebook.ipynb`

Key modules and results:

- Cell 1–20: Preprocessing, EDA, correlation analysis.
- Cell 21–32: Unsupervised learning (K-Means, hierarchical, PCA, t-SNE).
- Cell 33–50: Ensemble methods (Random Forest, XGBoost, Naive Bayes).
- Cell 51–72: Neural networks (SLP, MLP, ablation, CNN).
- Cell 73–76: Streamlit dashboard and model serialization.

Generated images are saved in:

`images/01_*.png` to `images/23_*.png`

Dashboard code is in:

`app/app.py` (Streamlit front-end)

Bibliography

[1] Dua, D., & Graff, C. (2019). UCI Machine Learning Repository. University of California, Irvine, School of Information and Computer Sciences.

[2] Arthur, D., & Vassilvitskii, S. (2007). k-means++: The advantages of careful seeding. In *Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms* (pp. 1027–1035).

[3] Van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9(11), 2579–2605.

[4] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794).

[5] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* (pp. 4765–4774).

[6] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *The Journal of Machine Learning Research*, 15(1), 1929–1958.

[7] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.

[8] Streamlit Inc. (2023). Streamlit: A faster way to build and share data apps. Retrieved from `https://streamlit.io`